

6(a)(i)	any three from: 1 <i>trp</i> operon regulatory gene / <i>trpR</i>, codes for an inactive repressor and <i>lac</i> operon regulatory gene / <i>lacI</i>, codes for active repressor ; 2 <i>lac</i> operon does not have an attenuator ; ora 3 <i>lac</i> operon has fewer structural genes or 3 vs 5 structural genes ; 4 <i>lac</i> operon uses, an inducer / (allo)lactose and <i>trp</i> operon uses, a repressor / tryptophan ; 5 (allo)lactose, binds / inactivates, repressor, so repressor, leaves / cannot bind to, operator and tryptophan, binds / activates, repressor, so repressor can bind to operator ; 6 (allo)lactose causes genes to be, transcribed / switched on / expressed and tryptophan causes genes to be not, transcribed / switched on / expressed ;	3
6(a)(ii)	any one from: 1 (share) one promoter ; 2 all, enzymes / proteins / products, work together ;	1
6(a)(iii)	<i>structural genes</i>: code for, enzymes / structural proteins / non-regulatory proteins / rRNA / tRNA ; <i>regulatory genes</i>: code for, proteins / products, that control, gene expression / transcription ; A code for, transcription factors / repressor proteins	2

Question	Answer	Marks
6(a)(iv)	any two from: 1 end-product inhibition / negative feedback / feedback inhibition ; 2 <u>tryptophan</u>, binds to / activates, repressor or <u>tryptophan</u> allows repressor to bind to operator ; 3 <u>tryptophan</u>, stops / reduces, <i>trpA</i> or gene transcription / <i>trpA</i> or gene expression / protein synthesis / tryptophan synthase being made ;	2
6(b)(i)	any four from: 1 (gibberellin) binds to receptor ; 2 ref. enzyme ; 3 (enzyme causes) DELLA breakdown ; 4 DELLA no longer, binds to / inhibits, transcription factor / PIF ; 5 (so) transcription factor / PIF / RNA polymerase, binds to, DNA / promoter ; 6 (growth) genes, switched on / expressed / transcribed ;	4
6(b)(ii)	mark first answer transcription factor(s) ;	1